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A PC Algorithm for Max-Linear Bayesian Networks

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ABSTRACT

Max-linear Bayesian networks (MLBNs) are a relatively recent class of structural equation models which arise when the random variables involved have heavy-tailed distributions. Unlike most directed graphical models, MLBNs are typically not faithful to d-separation and thus classical causal discovery algorithms such as the PC algorithm or greedy equivalence search can not be used to accurately recover the true graph structure. In this paper, we begin the study of constraint-based discovery algorithms for MLBNs given an oracle for testing conditional independence in the true, unknown graph. We show that if the oracle is given by the $*$-separation criteria in the true graph, then the PC algorithm remains consistent despite the presence of additional CI statements implied by $*$-separation. We also introduce a new causal discovery algorithm named `PCstar` which assumes faithfulness to C^*-separation and is able to orient additional edges which cannot be oriented with only d- or $*$-separation. Supplementary materials for this article are available online.

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1. Introduction

Structural equation models on directed acyclic graphs are often used to model the causal relationships between the components of a random vector $X = (X_1, \ldots, X_n)$. In a structural equation model, the component X_i of X is typically a function of its parents plus independent noise of the form $X_i = f_i(X_{\mathrm{pa}(i)}, \epsilon_i)$, where $\mathrm{pa}(i)$ is the set of parents of i. Structural equation models, sometimes called structural causal models or Bayesian networks, have been studied extensively over the past 30 years, especially in the context of the emerging field of *causality* (Pearl 2009), and have been used in a variety of applications including biology and economics (Durbin et al. 1998; Evans and Ringel 1999; Maathuis, Drton, and Lauritzen 2019). These models are particularly well understood in the case that the independent errors ϵ_i are Gaussian and the f_i are linear (Lauritzen 1996; Maathuis, Drton, and Lauritzen 2019), though significant progress has recently been made when the former assumption is dropped as well (Wang and Drton 2020; Tramontano, Drton, and Etesami 2026).

In this paper, we focus on a relatively new variation of structural causal models, known as *max-linear models* or max-linear Bayesian networks (MLBN) in which the random vector X follows the structural equation system

$$X_j = \bigvee_{i \in \mathrm{pa}(j)} c_{ij} X_i \vee Z_j, \quad c_{ij}, Z_j \geq 0, \tag{1.1}$$

where $\vee = \max$, the c_{ij} are edge weights, and the Z_j are independent, atom-free, continuous random variables. These models were first introduced in Gissibl and Klüppelberg (2018) to model scenarios with large risk, meaning that the variables Z_j are typically assumed to be heavy-tailed and thus often take extreme values. Max-linear models have since become one of the core approaches for modeling causal relationships in extreme scenarios alongside the parametric setting described in Engelke, Gnecco, and Röttger (2025). Max-linear models excel at modeling scenarios which involve cascading failure in which one extreme event causes other extreme events throughout the network as often seen in applications such as financial risk (Einmahl, Kiriliouk, and Segers 2018) or ecology (Asadi, Davison, and Engelke 2015).

One fundamental problem concerning any structural causal model is that of *causal discovery* in which the goal is to reconstruct a directed graph whose edges encode causal relationships between random variables using data. This problem has been studied extensively for most classical families of directed graphical models and has led to the development of many efficient causal discovery algorithms (Glymour, Zhang, and Spirtes 2019). These causal discovery algorithms generally fit into two paradigms known as *score-based* algorithms and *constraint-based* algorithms. Score-based algorithms such as Greedy Equivalence Search (GES) (Chickering 2003) attempt to optimize a score-function while moving between graphs using a restricted set of edge additions, deletions, and reversals. On the other hand, constraint-based algorithms such as the PC algorithm (Spirtes and Glymour 1991) attempt to remove and orient edges such that the resulting graph satisfies exactly the conditional independence statements which are present in the observed data. In almost all existing causal discovery algorithms, *faithfulness* to d-separation is required for consistency. Unlike most directed graphical models, MLBNs are not generically

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faithful to the well known d-separation criteria (Geiger, Verma, and Pearl 1990), meaning that a full dimensional subset of distributions in the model satisfy additional conditional independence (CI) statements which are not implied by d-separation on the underlying graph. However, Améndola et al. (2022) introduced the stronger criteria of $*$-separation and C^*-separation, which capture the additional CI statements which MLBNs satisfy and showed that MLBNs are generically faithful to the latter.

In this paper, we investigate constraint-based causal discovery algorithms for MLBNs. This means we assume that there exists a true underlying DAG $\mathcal{G}$ with weight matrix C and assume we have an oracle for testing conditional independence in the observed data. We first show that if the input to the PC algorithm is the global Markov property of $*$-separation on $\mathcal{G}$, that is the oracle produces the set of CI statements $\text{global}_*(\mathcal{G}) = \{[i \perp\!\!\!\perp j|K] \text{ s.t } [i\perp_* j|K] \text{ holds in } \mathcal{G}\}$, then the output will be identical to the result of running the PC algorithm with input $\text{global}_d(\mathcal{G})$. That is, despite the fact that $*$-separation yields additional CI statements, and thus MLBNs are not faithful to d-separation, the PC algorithm remains consistent and still produces the same result. We then turn our attention to constraint-based causal discovery with an oracle for checking C^*-separation in the pair $(\mathcal{G}, C)$ since this is the criteria to which MLBNs are actually faithful, and thus if the oracle were to be replaced with an actual conditional independence test, this would be the true input. We introduce a new algorithm named `PCstar` which takes as input an oracle for testing C^*-separation in the true unknown pair $(\mathcal{G}, C)$ and returns a partially oriented graph, $\mathcal{H}$,which faithfully encodes all conditional independencies yielded by the oracle. In particular, there exists an orientation $\mathcal{H}'$ of $\mathcal{H}$ and a positive dimensional set of coefficient matrices on $\mathcal{H}'$ such that $(\mathcal{H}', C_{\mathcal{H}'})$ and $(\mathcal{G}, C_{\mathcal{G}})$ yield the same set of statements under C^*-separation. Our algorithm is directly inspired by the PC algorithm and proceeds in three steps, which are *skeleton retrieval*, *collider detection*, and *cycle orientation*. Our first two steps are identical to that of the PC algorithm, albeit with the assumption of faithfulness to C^*-separation instead of d-separation. We then show how the additional statements which come from C^*-separation can be used to orient additional edges which lie in certain cycles. Lastly, we show that similarly to the PC algorithm, there exists a variant of our algorithm which is polynomial if the maximum in-degree of any node in the true graph is bounded. We end by showcasing how our algorithm performs on sparse random graphs.

The remainder of this paper is organized as follows. In Section 2 we provide background on max-linear models, separation criteria, and the PC algorithm. In Section 3, we show that the PC algorithm returns the same output when given an oracle for $*$-separation as it does for d-separation oracle despite the fact that the $*$-separation oracle applied to the unknown true graph yields additional CI statements. In Section 4, we introduce our new causal discovery algorithm for max-linear models, named `PCstar` and show that the output faithfully encodes all CI statements produced by the oracle. We then show that when the maximum in-degree of the true DAG $\mathcal{G}$, is bounded, `PCstar` is polynomial in the number of nodes n and that the extra statements which come from C^*-separation can often be used to orient additional edges which cannot be oriented under the assumption of faithfulness to d-separation. In Section 5, we show how `PCstar` performs on sparse random graphs and discuss some directions for future work.

2. Preliminaries

2.1. Graph Terminology

A *simple directed graph* $\mathcal{G} = (V, E)$ consists of nodes V and directed edges E, that is, edges of the form $i \to j$ such that there is at most one edge between any two nodes. We say that two nodes i and j are *adjacent* if they are connected by an edge. The set of all nodes adjacent to i are the *neighbors* of i, denoted $\text{ne}(i)$. An $i - j$ *path* π is a sequence of distinct nodes $(i, \dots, j)$ such that all consecutive nodes in π are adjacent. We say $\pi = (k_0 = i, \dots, j = k_s)$ is *directed* if all edges on π are of the form $k_i \to k_{i+1}$, i.e, point toward j. An $i - j$ path π is called a *trek* if there exists a $k \in \pi$ such that the subpaths $(i, \dots k)$ and $(k, \dots, j)$ are directed paths from k to i and k to j respectively. A directed path from i to j and the edge $j \to i$ form a *directed cycle*. We call a simple directed graph without directed cycles a *directed acyclic graph* (DAG). Given an edge $i \to j$ we say that i is a *parent* of j and denote the set of parents of j with $\text{pa}(j)$. The cardinality of $\text{pa}(j)$ is the *in-degree* of j, denoted $\text{indeg}(j)$. Given a directed path $i \to \cdots \to j$ we say that i is an *ancestor* of j, respectively j is a *descendant* of i. We denote the set of ancestors with $\text{an}(j)$ and the set of descendants with $\text{de}(i)$. For a subset of nodes $K \subset V$ we define $\text{an}(K) := \cup_{k\in K}\text{an}(k)\backslash K$ and $\text{An}(K) := \text{an}(K) \cup K$. The sets $\text{de}(K)$ and $\text{De}(K)$ are defined analogously.

The *undirected skeleton* of a directed graph $\mathcal{G} = (V, E)$ is the undirected graph with node set V containing the edge $i - j$ if and only if i and j are adjacent in $\mathcal{G}$, and is denoted $\text{skel}(\mathcal{G})$. The notions of path, neighbors, and cycles in undirected graphs are defined analogously.

2.2. Separation Criteria

Graphical models identify conditional independence relations through a separation condition $\perp$. The best-known instance of this in directed acyclic graphical models is *d-separation* for linear structural equation models. A triple of nodes $\{i, k, j\} \subset V$ form a *collider* in a DAG $\mathcal{G} = (V, E)$ if $i \to k \in E$ and $j \to k \in E$. The collider is said to be *unshielded* if $i \to j \notin E$ and $j \to i \notin E$. An $i - j$ path π in a DAG is *d-connecting* given $K \subset V\backslash ij$ if all colliders along π lie in $\text{An}(K)$ and no non-collider along π lies in K. If a d-connecting path exists, we say that i and j are *d-connected* given K. Otherwise, we say that they are *d-separated* and write $[i\perp_{\text{d}} j|K]$.

It is well-known (Geiger, Verma, and Pearl 1990, Theorem 2) that linear structural equation models on a DAG $\mathcal{G}$ are *faithful* to d-separation in the sense that for all $i, j, K \subset V$:

$$[X_i \perp\!\!\!\perp X_j|X_K] \text{ holds in } X \iff [i\perp_{\text{d}} j|K] \text{ holds in } \mathcal{G}. \quad (2.1)$$

Equivalently, the complete set of conditional independence statements in the model can be recovered from its *Global Markov property*, defined as

$$\text{global}_d(\mathcal{G}) = \{[i \perp\!\!\!\perp j|K \text{ s.t } [i\perp_{\text{d}} j|K] \text{ holds in } \mathcal{G}\}, \quad (2.2)$$

Algorithm 1 Skeleton Learning Step of the PC Algorithm

Input : A complete set of CI statements $\text{global}_{\text{d}}(\mathcal{G})$ coming from a graphical model faithful to d-separation
Output: The undirected skeleton of $\mathcal{G}$

1 Set G to be the complete graph on n nodes
2 **for** ℓ *from* 0 *to* $\max_{v \in V(\mathcal{G})} \text{indeg}(v)$ **do**
3 **for** *each edge* $(i,j) \in G$ **do**
4 **for** *each* $K \subseteq \text{ne}(i) \cup \text{ne}(j) \setminus \{i,j\}$ *s.t.* $|K| = \ell$ **do**
5 **if** $(i,k) \in G$ *for all* $k \in K$ ***or*** $(j,k) \in G$ *for all* $k \in K$ **then**
6 **if** $[i \perp\!\!\!\perp j \mid K] \in \text{global}_{\text{d}}(\mathcal{G})$ **then**
7 remove edge (i,j) from G
8 Set $Sepset(i,j) := K$

where we implicitly identify X_i, X_j and X_K with i, j, and K respectively.

The faithfulness property is what allows the reconstruction of all of the probabilistic CI statements in a distribution from the purely combinatorial properties of $\mathcal{G}$ and vice versa. Two graphs $\mathcal{G}$ and $\mathcal{H}$ are said to be *Markov equivalent* with respect to a d-separation if $\text{global}_d(\mathcal{G}) = \text{global}_d(\mathcal{H})$. We write this as $\mathcal{G} \sim_d \mathcal{H}$. This relation partitions the set of DAGs on n nodes in the following way:

Theorem 2.1 (Andersson, Madigan, and Perlman 1997, *Theorem 2.1). Two DAGs $\mathcal{G}$ and $\mathcal{H}$ with equal node set are d-Markov equivalent if and only if they have the same undirected skeleton and unshielded colliders.*

2.3. Causal Discovery and the PC Algorithm

The term *causal discovery* is used to refer to the following task:

> *Given data which is known to come from a graphical model $(\mathcal{G}, X)$, recover$\mathcal{G}$*

The PC algorithm of Spirtes and Glymour (1991) is a constraint-based algorithm which outputs a graph approximating the true DAG $\mathcal{G}$ from conditional independence statements. We describe the two steps of the algorithm under the idealized assumption that there exists a perfect test for asserting whether a given conditional independence statement $[i \perp\!\!\!\perp j|K]$ lies in $\text{global}_{\text{d}}(\mathcal{G})$ (or equivalently, that the entire set $\text{global}_{\text{d}}(\mathcal{G})$ is known). The first step is the *skeleton learning step*, which retrieves the undirected skeleton of the true DAG $G := \text{skel}(\mathcal{G})$ by querying the set $\text{global}_{\text{d}}(\mathcal{G})$.

A proof of correctness of Algorithm 1 may be found in Geiger, Verma, and Pearl (1990). An important feature is present on line 2; in order to retrieve the adjacencies of the true DAG $\mathcal{G}$ it is sufficient to query $\text{global}_{\text{d}}(\mathcal{G})$ for CI statements $[i \perp\!\!\!\perp j|K]$ in which the size of K is bounded by the maximal in-degree of $\mathcal{G}$. This is because any two non-adjacent nodes $i, j \in V$ are d-separated by a subset of either $\text{pa}(i)$ or $\text{pa}(j)$ (Verma and Pearl 2022, Lemma 1). In particular, this ensures that the worst case performance of Algorithm 1 is n^{a+2}, where a is the maximal in-degree of a node in $\mathcal{G}$ (see (Meek 2023, p. 138).

After completing the skeleton learning step, the PC algorithm iterates through the edges of the undirected skeleton of $\mathcal{G}$ and orients them according to the four rules stated in List 1. We refer to this as the *edge orientation step* of the PC algorithm. More details may be found in Meek (2023). It makes use of the sets $Sepset(i,j)$ collected in Algorithm 1, which for each non-adjacent pair $\{i,j\}$ record a set K of minimal cardinality such that $[i \perp\!\!\!\perp j|K] \in \text{global}_{\text{d}}(\mathcal{G})$. Specifically, the PC algorithm first iterates through the *unshielded triples* of G (i.e. triples of nodes $\{i,j,k\}$ such that $i - j$ and $j - k$ are edges of G, but $i - k$ is not) and orients them according to $\mathcal{R}1$. It then proceeds to apply the remaining rules, which are the only choices of edge directions which do not create directed cycles or additional unshielded colliders in the graph. The complexity of this procedure lies in $\mathcal{O}(n^3)$ (Meek 2023, pg.143), hence the total runtime of the algorithm remains bounded above by n^{a+2}.

$\mathcal{R}1$: For any unshielded triple $i - k - j$, orient the edges as $i \to k \leftarrow j$ if $k \notin Sepset(i,j)$ holds.
$\mathcal{R}2$: For any remaining unshielded triple oriented $i \to k - j$, orient it as $i \to k \to j$.
$\mathcal{R}3$: For any shielded triple where $i \to k \to j$, orient $i - j$ as $i \to j$.
$\mathcal{R}4$: For any four-node diamond structure' such as the one depicted on the right in Figure 1, orient $i - \ell$ as $i \to \ell$ if $j - \ell - k$ is oriented $j \to \ell \leftarrow k$.
List 1: The four edge orientation rules of the PC algorithm.

The output of the edge orientation step is a *complete partially oriented directed acyclic graph*, or CPDAG (see Spirtes, Glymour, and Scheines 1993, Chapter 5). It has the same unshielded colliders as the true DAG $\mathcal{G}$ and thus represents the Markov equivalence class of $\mathcal{G}$ due to Theorem 2.1. It is the best approximation of $\mathcal{G}$ that can be obtained from the conditional independence information of a linear structural equation model on $\mathcal{G}$ without additional interventional data (Meek 1995, Theorem 3).

2.4. Star Separation Criteria for MLBNs

Max-Linear Bayesian networks are generally not faithful to d-separation; more precisely, additional conditional independence may hold in the model which is not implied by d-separation.

This motivated the introduction of stronger separation conditions for Max-Linear Bayesian Networks. This was done in Améndola et al. (2022). We recall the relevant notions and definitions of both weighted and unweighted *star separation*.

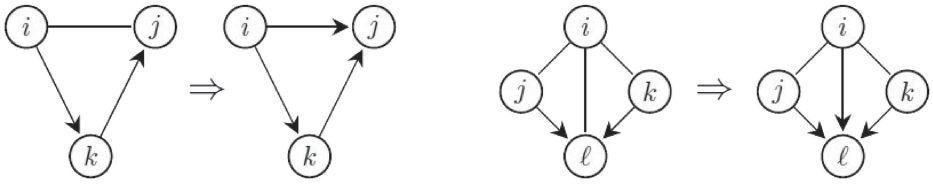


Figure 1. List 1: The four edge orientation rules of the PC algorithm.

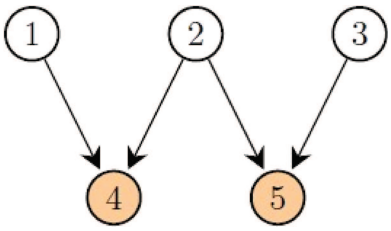


Figure 2. The Cassiopeia graph.

2.5. Unweighted ∗-Separation

We adopt the characterization of $\perp_*$ which was developed in Améndola, Hollering, and Sullivant (2021). An undirected $i - j$ path in a DAG $\mathcal{G}$ is $*$-*connecting* given $K \subset V \backslash ij$ if it is d-connecting and contains at most one collider. If no such path in $\mathcal{G}$ exists, we say that i and j are $*$-*separated* given K, and denote this by $[i \perp_* j | K]$.

Example 2.2 (Améndola et al. 2022, Example 4.2). In Figure 2, the nodes 1 and 2 are $*$-connected given $K = \{4, 5\}$. 1 and 3 are $*-$ separated given the same K. In the max-linear model on this DAG for any choice of coefficients c_{ij}, the CI statement $[1 \perp\!\!\!\perp 3|45]$ holds. To see this in depth, consider the case in which all edge weights c_{ij} are equal to one. If one conditions on $X_4 = x_4$ and $X_5 = x_5$ for $x_4, x_5 \in \mathbb{R}$, then the max-linear equations imply that $X_1 \vee X_2 \leq x_4$ and $X_2 \vee X_3 \leq x_5$. If $x_4 < x_5$ holds, then combining the inequalities gives $X_2 < x_5$, implying that X_2 cannot be the true cause of x_5. Similarly, if $x_4 > x_5$, then x_4 can only be caused by X_1 or Z_4. Finally, if $x_4 = x_5$, then the fact that the Z_i are atom-free imply that X_2 is almost surely the true common cause of X_4 and X_5. In all three of these cases, X_1 and X_3 are independent, hence the statement $[X_1 \perp\!\!\!\perp X_3 | X_{\{4,5\}}]$.

From the definition of $\perp_*$ it is clear that $[i \perp_{\mathrm{d}} j | K]$ implies $[i \perp_* j | K]$ for all *i, j, K*. On the level of the global Markov properties as in (2.2), this means that

$$\mathrm{global}_{\mathrm{d}}(\mathcal{G}) \subset \mathrm{global}_*(\mathcal{G}). \tag{2.3}$$

Example 2.2 is an instance where strict inclusion in (2.3) holds. Despite this phenomenon the corresponding relations $\sim_d$ and $\sim_*$ partition the set of all DAGs into the same Markov equivalence classes.

Theorem 2.3 (Améndola, Hollering, and Sullivant 2021, *Theorem 3.4). For any two DAGs $\mathcal{G}$ and $\mathcal{H}$: $\mathcal{H} \sim_{\mathrm{d}} \mathcal{G} \iff \mathcal{H} \sim_* \mathcal{G}$.*

The behavior of the PC algorithm under $\perp_*$ is investigated in Section 3.

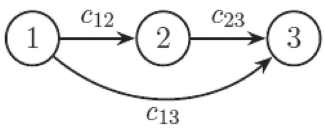


Figure 3. A complete weighted DAG on 3 nodes.

2.6. The Weighted Case: C*-Separation

We now consider the more general setting of *weighted* Max-Linear Bayesian Networks. Let $\mathcal{G}$ be a DAG on n nodes with edge set E and denote the set of coefficient matrices supported on $\mathcal{G}$ by $\mathbb{R}^E$; this is the set of all $n \times n$ matrices C with $c_{ij} = 0$ if $i \to j \notin E$ and $c_{ij} \in \mathbb{R}_{>0}$ otherwise. We call a pair $(\mathcal{G}, C)$ a *weighted DAG*. A random vector X is distributed according to the MLBN on $(\mathcal{G}, C)$ if it fulfills (1.1). The introduction of edge weights brings with it additional challenges, which we illustrate with two examples. Firstly, the set of CI statements which hold in a weighted MLBN depend on the choice of weight matrix C.

Example 2.4. Consider the complete weighted DAG $(\mathcal{G}, C)$ on 3 nodes (Figure 3).

The sets of CI statements which hold in the corresponding MLBN are

$$\begin{cases} \{[X_1 \perp\!\!\!\perp X_3 | X_2]\} & \text{if } c_{13} \leq c_{12}c_{23} \\ \emptyset & \text{if } c_{13} > c_{12}c_{23}. \end{cases}$$

The conditional independence statement in the first case arises due to the fact that if the inequality $c_{13} \leq c_{12}c_{23}$ holds, the structural equation for X_3 can be rewritten as

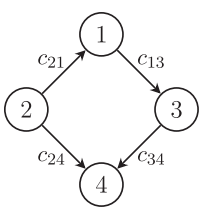
$$\begin{aligned} X_3 &= c_{13}X_1 \vee c_{23}X_2 \vee Z_3 \\ &= c_{13}X_1 \vee c_{23}(c_{12}X_1 \vee Z_2) \vee Z_3 \\ &= (c_{13} \vee c_{12}c_{23})X_1 \vee c_{23}Z_2 \vee Z_3 \\ &= c_{12}c_{23}X_1 \vee c_{23}Z_2 \vee Z_3 = c_{23}X_2 \vee Z_3. \end{aligned}$$

In particular, X_1 influences X_3 only indirectly via X_2. □

The following example shows weighted MLBNs are not faithful to the criterion $\perp_*$.

Example 2.5. Consider the graph depicted below with edge weights chosen such that

$c_{24} < c_{21}c_{13}c_{34}$ and corresponding structural equations:



$$X_1 = c_{21}X_2 \vee Z_1$$

$$X_2 = Z_2$$

$$X_3 = c_{13}X_1 \vee Z_3 = c_{13}(c_{21}Z_2 \vee Z_1) \vee Z_3$$

$$X_4 = c_{24}X_2 \vee c_{34}X_3 \vee Z_4$$

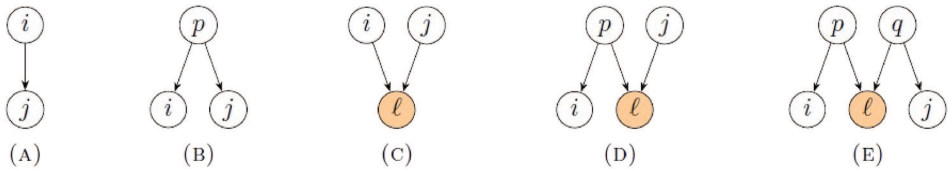


Figure 4. The types of $*$-connecting paths between i and j given K in a critical DAG $\mathcal{G}_K^*$. The colored colliders ℓ must belong to K; the non-colliders p, q must not belong to K.

In particular, we observe that the conditional independence statement $[X_2 \perp\!\!\!\perp X_4|X_3]$ holds in this model. However, $[2 \perp\!\!\!\perp 4|3]$ is not a statement in $\text{global}_*(\mathcal{G})$ as the edge $2 \to 4$ is a $*-$connecting path given $K = \{3\}$. □

Intuitively, the reason for the additional CI statement in the example above is that, due to this particular choice of edge weights, the edge $2 \to 4$ is no longer *critical*; the random variable X_2 influences X_4 only indirectly via the path $2 \to 1 \to 3 \to 4$. Phenomena such as this one motivate yet another notion of graph separation which can account for critical paths. This was introduced in Améndola et al. (2022). Let $(\mathcal{G}, C)$ be a weighted DAG on n nodes and $i, j \in V(\mathcal{G})$. We denote the set of all directed paths from i to j by $P(i, j)$. For a directed path $\pi = i \to l_1 \to \cdots \to l_m \to j \in P(i, j)$, we define its *weight* in $(\mathcal{G}, C)$ as $\omega_C(\pi) = c_{il_1}c_{l_1l_2} \cdot \cdots \cdot c_{l_mj}$. We say that a directed path $\pi \in P(i, j)$ is *critical* if its weight $\omega_C(\pi)$ attains the maximal value over all $P(i, j)$. The weights of the critical paths of $(\mathcal{G}, C)$ are encoded in the *Kleene star* of C. This is the matrix

$$C^* = (c_{ij}^*)_{i,j\in[n]} = \max_{\pi\in P(i,j)} \omega_C(\pi),$$

which may be computed by taking a tropical power of C (Butkovič 2010, Proposition 1.6.15).

To define a notion of graph separation for weighted DAGs we associate a graph to each conditioning set K. We say that a directed path π from i to j *factors through* K if there exists a node k along π such that $k \in K\backslash ij$. Let $\mathcal{G}$ be a DAG on n nodes and $K \subset V$. The *critical DAG* of $(\mathcal{G}, C)$ with respect to K, denoted $\mathcal{G}_K^*(C)$ is the unweighted DAG with node set V and edges determined by the condition

$$i \to j \in \mathcal{G}_K^*(C) \iff c_{ij}^* > 0 \text{ and no critical directed path from } i \text{ to } j \text{ factors through } K.$$

If C is fixed and clear from context, we may write $\mathcal{G}_K^*$ for readability. We say that two nodes i and j in $V\backslash K$ are C^*-*connected* given K if there is a path from i to j in $\mathcal{G}_K^*$ of one of the five types (A)–(E) depicted in Figure 4. If no such path in $\mathcal{G}_K^*$ exists, we say that i and j are C^**-separated* given K, and denote this by $[i\perp_{C^*}j|K]$.

It was proven in Améndola et al. (2022) that for all but a zero-measure set of weight matrices in $\mathbb{R}^E$, weighted MLBNs are faithful to the $\perp_{C^*}$ condition in the sense that

$$[i\perp_{C^*}j|K] \text{ holds in } (\mathcal{G}, C) \Leftrightarrow [X_i \perp\!\!\!\perp X_j|X_K] \text{ holds in } X, \quad (2.4)$$

where $X = (X_1, \ldots X_n)$ is a MLBN on $(\mathcal{G}, C)$. Equivalently, the *global Markov property* of a weighted DAG $(\mathcal{G}, C)$ w.r.t. $\perp_{C^*}$

$$\text{global}_{C^*}(\mathcal{G}, C) = \{[i \perp\!\!\!\perp j|K] \text{ s.t } [i\perp_{C^*}j|K] \text{ holds in } (\mathcal{G}, C)\}$$

encodes the entire conditional independence structure of its corresponding MLBN.

Proposition 2.6. Let $\mathcal{G}$ be a DAG. For any coefficient matrix $C \in \mathbb{R}^E$ it holds that

$$\text{global}_{\text{d}}(\mathcal{G}) \subset \text{global}_*(\mathcal{G}) \subset \text{global}_{C^*}(\mathcal{G}, C). \quad (2.5)$$

This is a reformulation of Corollary 5.9 of Améndola et al. (2022). Example 2.2 and Example 2.5 are instances of strict inclusion in the first and second inclusions respectively; the latter fact will be shown in Example 4.1.

3. The PC Algorithm in Unweighted MLBNs

The goal of this section is to prove the following statement:

Theorem 3.1. Let $\mathcal{G}$ be a DAG, and $\text{global}_{\text{d}}(\mathcal{G})$ and $\text{global}_*(\mathcal{G})$ be its global Markov properties with respect to $\perp_{\text{d}}$ resp. $\perp_*$. Applying the PC algorithm to $\text{global}_*(\mathcal{G})$ returns the same CPDAG as the one obtained upon taking $\text{global}_{\text{d}}(\mathcal{G})$ as input.

By Theorem 2.3 this is a desirable result, as a graph with the same skeleton and unshielded colliders as $\mathcal{G}$ is a representative of the $\perp_*$ Markov equivalence class of $\mathcal{G}$. The first three results of this section concern the skeleton retrieval step. We say that two nodes i and j are *separated* with respect to a separation condition $\perp$ if $[i\perp j|K]$ holds for some $K \subset V\backslash ij$.

Lemma 3.2. Let $\mathcal{G} = (V, E)$ be a DAG and $i, j \in V$. The following are equivalent:

(1) i and j are non-adjacent.
(2) i and j are d-separated.
(3) i and j are $*$-separated.

Proof. $1 \Rightarrow 2$ is the "implied" direction of (Verma and Pearl 2022, Lemma 1), whereas $2 \Rightarrow 3$ follows directly from the definition of $*$-separation. $3 \Rightarrow 1$ is easily proven by contraposition: if i and j are adjacent, then the edge between them is a $*-$connecting path for any choice of $K \subset V\backslash ij$. □

For the correctness of the skeleton retrieval step, in addition to Theorem 3.2 one has to show that, similarly to the d-separation setting, it suffices to consider statements in $\text{global}_*(\mathcal{G})$ where the size of K is bounded in order to recover the correct skeleton.

Lemma 3.3. Let i and j be non-adjacent nodes in a DAG $\mathcal{G}$. Then either:

$$[i\perp_* j|\text{pa}(j)] \quad \textit{or} \quad [i\perp_* j|\text{pa}(i)].$$

Proof. Due to the acyclicity of $\mathcal{G}$, we may assume without loss of generality that there exists no directed path from i to j. We claim that in this case, $[i \perp_* j | \mathrm{pa}(i)]$ holds. We do this by showing that no undirected path between i and j is $*-$ connecting given $\mathrm{pa}(i)$. Paths between i and j containing no colliders are either directed paths $i \leftarrow \cdots \leftarrow j$ or treks of the form $i \leftarrow \cdots \leftrightarrow \cdots \rightarrow j$. Both of these types of path are blocked by $\mathrm{pa}(i)$. Paths between i and j which contain exactly one collider and are not blocked by $\mathrm{pa}(i)$ are of the form $i \rightarrow \cdots \rightarrow k_1 \rightarrow k \leftarrow k_2 - \cdots - j$, where the portion of the path from k_2 to j is either directed or a trek. For this path to be $*$- connecting given $\mathrm{pa}(i)$, the middle node of the collider k would have to be an ancestor of a node in $\mathrm{pa}(i)$, and thus also of i. However, this would imply that $\mathcal{G}$ contains the directed cycle $i \rightarrow \cdots \rightarrow k \rightarrow \cdots \rightarrow i$, contradicting the assumption that $\mathcal{G}$ is acyclic. □

Proposition 3.4. The output of Algorithm 1 does not change upon replacing $\mathrm{global}_{\mathrm{d}}(\mathcal{G})$ *with* $\mathrm{global}_{*}(\mathcal{G})$ *as input.*

Proof. Any two non-adjacent nodes i, j in $G = \mathrm{skel}(\mathcal{G})$ are separated by a conditioning set of cardinality at most $\max\{|\mathrm{pa}(i)|, |\mathrm{pa}(j)|\}$ due to a combination of Lemma 3.2 and Lemma 3.3. The fact that Lemma 3.2 is an equivalence means that no additional edges are cut when applying Algorithm 1 to $\mathrm{global}_{*}(\mathcal{G})$. Hence, the output is once again that of Algorithm 1, which is the undirected skeleton G. □

Finally, we need to show that the edge orientation step behaves the same in both settings.

Lemma 3.5. The edge orientation step of PC does not change upon replacing $\mathrm{global}_{\mathrm{d}}(\mathcal{G})$ with $\mathrm{global}_{*}(\mathcal{G})$ as input.

Proof. The triples oriented by $\mathcal{R}1$ of List 1 depend entirely on conditional independence statements with $|K| = 1$. Subsequent edge orientations according to the rules $\mathcal{R}2 - \mathcal{R}4$ are not dependent on the statements taken as input. Thus, the claim follows from the fact that if $|K| \leq 1$, then $[i \perp_{\mathrm{d}} j | K] \iff [i \perp_* j | K]$ holds. The implication from left to right follows from (2.3). For the converse implication, we prove the contrapositive. Let π be a d-connecting $i - j$ path, conditioned on $K = \{k\}$. The path π is of the form $\pi : i - \cdots \rightarrow a_1 \leftarrow \cdots \rightarrow a_m \leftarrow \cdots - j$, where $a_1, \ldots, a_m \in \mathrm{an}(K)$ are the conditioned-on colliders. If $m \in \{0, 1\}$, then π is $*$-connecting by definition. If $m \geq 2$, then the path $\pi' : i - \cdots a_1 \rightarrow \cdots \rightarrow k \leftarrow \cdots \leftarrow a_m - \cdots - j$, obtained by concatenating the $i - a_1$ and $a_m - j$ portions of π with the path $a_1 \rightarrow \cdots \rightarrow k \leftarrow \cdots \leftarrow a_m$ is a $*$-connecting path, implying $[i \not\perp_* j | K]$ as desired. □

The proof of Theorem 3.1 now follows by combining Proposition 3.4 and Lemma 3.5.

4. The PCstar Algorithm for Weighted MLBNs

In this section we discuss the behavior of the PC algorithm when the input is a set of CI statements $\mathrm{global}_{C^*}(\mathcal{G}, C)$ arising from a weighted MLBN. In this setting, the additional CI statements from (2.6) lead Algorithm 1 to output a CPDAG whose undirected skeleton may no longer be equal to the undirected skeleton of $\mathcal{G}$.

Example 4.1. Assume that the true DAG $(\mathcal{G}, C)$ underlying a MLBN is the graph depicted in with coefficient matrix C chosen such that $c_{24} < c_{21}c_{13}c_{34}$. The set $\mathrm{global}_{C^*}(\mathcal{G}, C)$ is

$$[1 \perp\!\!\!\perp 4|3],\ [1 \perp\!\!\!\perp 4|23]$$
$$[2 \perp\!\!\!\perp 3|1],\ [2 \perp\!\!\!\perp 3|14]$$
$$[2 \perp\!\!\!\perp 4|1],\ [2 \perp\!\!\!\perp 4|3],\ [2 \perp\!\!\!\perp 4|13].$$

Applying Algorithm 1 to $\mathrm{global}_{C^*}(\mathcal{G}, C)$ outputs the subgraph depicted in Figure 5:

Unlike the $\perp_{\mathrm{d}}$ and $\perp_*$ cases, this is a proper subgraph of the undirected skeleton of $\mathcal{G}$. □

To make sense of this phenomenon, we recall the notion of the *weighted transitive reduction*.

4.1. Skeleton Retrieval and the Weighted Transitive Reduction

Definition 4.2. Let $(\mathcal{G}, C)$ be a weighted DAG on n nodes. The *weighted transitive reduction* of $(\mathcal{G}, C)$ is the weighted DAG $(\mathcal{G}_C^{tr}, C^{tr})$ on n nodes with weighted edges determined as follows:

$$i \rightarrow j \in \mathcal{G}_C^{tr} \text{ with weight } c_{ij} \ :\iff \text{The edge } i \rightarrow j \text{ is the unique critical path from } i \text{ to } j \text{ in } \mathcal{G}.$$

Clearly, $\mathcal{G}_C^{tr}$ is a subgraph of $\mathcal{G}$. Versions of it appear in Gissibl and Klüppelberg (2018) (under the name *minimum max-linear DAG*) and in Améndola and Ferry (2026) where it is used in the combinatorial classification of polytropes. The results from this section will show that it is the sparsest subgraph of $\mathcal{G}$ which preserves its global Markov property.

Definition 4.3. For a fixed DAG $\mathcal{G}$ with edge set E, a matrix $C \in \mathbb{R}^E$ is *generic* if it is such that for all $i, j \in V$, the critical $i - j$ path in $(\mathcal{G}, C)$ is unique if it exists.

The term generic is justified by the fact that the set of all matrices supported on a fixed DAG $\mathcal{G}$ which give rise to multiple critical $i-j$ paths form a zero-measure set in $\mathbb{R}^E_{>0}$. Recent work of Boege et al. (2025) associates each DAG $\mathcal{G}$ to a polyhedral fan in the space of weight matrices $\mathbb{R}^E_{>0}$. In this fan, these zero-measure sets are precisely the faces of codimension ≥ 1.

Remark 4.4. For generic C, $\mathcal{G}_C^{tr}$ is the unique minimal subgraph which has the same critical paths as $\mathcal{G}$. This is because in any critical path $i \rightarrow \cdots k_1 \rightarrow k_2 \cdots \rightarrow j$ in $\mathcal{G}$, each intermediate edge $k_1 \rightarrow k_2$ is the unique critical $k_1 - k_2$ path in $\mathcal{G}$, and the property of being a critical path is transitive.

Two graphs with equal unique critical paths for each pair (i, j) give rise to the same global Markov properties. For a two nodes i, j in a weighted DAG $(\mathcal{G}, C)$ with generic C, we denote the sequence of nodes $(i, k_1, k_2, \ldots, j)$ corresponding to the unique critical $i - j$ path by $\pi^{ij}_{\mathrm{crit}}(\mathcal{G}, C)$.

Lemma 4.5 (Boege et al. 2025, *Lemma 2.4). Let* $(\mathcal{G}, C)$ *and* $(\mathcal{G}', C')$ *be two weighted DAGs on n nodes with generic weights C and C′. Then*

$$\mathrm{global}_{C^*}(\mathcal{G}, C) = \mathrm{global}_{C^*}(\mathcal{G}', C')$$

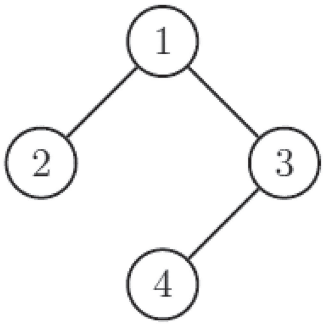


Figure 5. The output of the skeleton retrieval step of PC on the graph from is the complete undirected graph with the edges {1, 4}, {2, 3}, {2, 4} cut.

$$\iff \pi^{ij}_{\text{crit}}(\mathcal{G}, C) = \pi^{ij}_{\text{crit}}(\mathcal{G}', C') \quad \text{for all } i \neq j.$$

Corollary 4.6. For generic $(\mathcal{G}, C)$, $\mathcal{G}^{tr}_C$ *is the unique sparsest subgraph of* $\mathcal{G}$ *capable of encoding* $\text{global}_{C^*}(\mathcal{G}, C)$*, in the sense that it fulfills the following condition:*

$$\begin{gathered} \textit{There exists a } C' \textit{ supported on } \mathcal{G}^{tr}_C \textit{ such that} \\ \text{global}_{C^*}(\mathcal{G}, C) = \text{global}_{C^*}(\mathcal{G}^{tr}_C, C'). \end{gathered} \tag{4.1}$$

Proof. The equality $\text{global}_{C^*}(\mathcal{G}, C) = \text{global}_{C^*}(\mathcal{G}^{tr}_C, C^{tr})$ follows from the previous lemma and Remark 4.4. It is evident from Definition 4.2 that deleting any edge of $\mathcal{G}^{tr}_C$ would change its critical paths, thus implying that it is the sparsest subgraph of $\mathcal{G}$ fulfilling (4.1). □

The following result characterizes the behavior of the skeleton retrieval step of PC in the weighted setting (Figure 5).

Lemma 4.7. Let $(\mathcal{G}, C)$ *be a weighted DAG. For* $i, j \in V$ *it holds that*

$$[i \perp_{C^*} j | \text{pa}(j)] \text{ in } (\mathcal{G}, C) \iff [i \perp_* j | \text{pa}(j)] \text{ in } \mathcal{G}^{tr}_C.$$

Proof. We assume that there exists a directed path from i to j (In all other cases the equivalence is trivial).

"⇒": If $[i \perp_{C^*} j | \text{pa}(j)]$ holds in $\mathcal{G}$ then in particular $i \to j$ is not an edge of $\mathcal{G}^*_{\text{pa}(j)}$ (There is no $*$-connecting path of type (A) in this critical DAG). Thus, there exists a critical path from i to j which factors through $\text{pa}(j)$. This implies $i \to j \notin \mathcal{G}^{tr}_C$, and therefore (by Lemma 3.2) i and j are $*-$ separated by $\text{pa}(j)$ in $\mathcal{G}^{tr}_C$.

"⇐": If $[i \perp_* j | \text{pa}(j)]$ holds in $\mathcal{G}^{tr}_C$, then this implies $[i \perp_{C^*} j | \text{pa}(j)]$ holds in $\mathcal{G}^{tr}_C$ by Proposition 2.6. Due to Lemma 4.5, the statement also holds in $(\mathcal{G}, C)$. □

Theorem 4.8. Let $(\mathcal{G}, C)$ *be a weighted DAG. Applying Algorithm 1 to* $\text{global}_{C^*}(\mathcal{G}, C)$ *retrieves* $\text{skel}(\mathcal{G}^{tr}_C)$.

Proof. Let $i, j \in V$ and assume without loss of generality that no directed path from j to i exists. Using Lemma 3.2 and the definition of the global Markov property, Lemma 4.7 can be reformulated as follows:

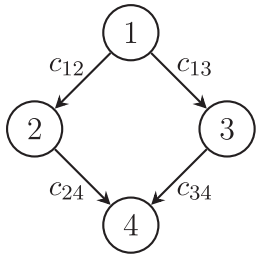
$$\begin{aligned} & i \text{ and } j \text{ are non} - \text{adjacent in } \mathcal{G}^{tr}_C \\ \iff & [i \perp_{C^*} j | \text{pa}(j)] \text{ holds in } (\mathcal{G}, C) \\ \iff & [i \perp\!\!\!\perp j | K] \in \text{global}_{C^*}(\mathcal{G}, C). \end{aligned}$$

Therefore, the procedure of starting with the complete undirected graph and removing edges whenever two nodes are C^*-separated recovers $\text{skel}(\mathcal{G}^{tr}_C)$. Furthermore (similarly to the d- and unweighted $*-$-separation cases) it suffices to query only the separation statements with conditioning set of cardinality up to $\max_{v \in V} \text{indeg}(v)$ due to Lemma 4.7. □

4.2. Collider Orientation

Applying Algorithm 1 to $\text{global}_{C^*}(\mathcal{G}, C)$ yields the skeleton of a well-defined subgraph of $(\mathcal{G}, C)$ which contains all of its critical paths. As $G^{tr} := \text{skel}(\mathcal{G}^{tr}_C)$ is generally a proper subgraph of the skeleton of $\mathcal{G}$, it may not contain all of its unshielded triples. Nevertheless, given $\text{global}_{C^*}(\mathcal{G}, C)$ one may expect to be able to orient some of the edges of G^{tr} using the rules in List 1, starting with $\mathcal{R}1$. The following example illustrates that a direct application of $\mathcal{R}1$ can lead to an incorrect output.

Example 4.9. Consider the weighted DAG $(\mathcal{G}, C)$ depicted below



with edges weighted such that the path $1 - 3 - 4$ is the unique critical $1 - 4$ path (i.e. $c_{12}c_{24} < c_{13}c_{34}$). The global Markov property is

$$\text{global}_{C^*}(\mathcal{G}, C) = \{[1 \perp\!\!\!\perp 4 \,|3]\,,\ [1 \perp\!\!\!\perp 4 \,|23]\,,\ [2 \perp\!\!\!\perp 3 \,|1]\}.$$

Applying the skeleton retrieval of PC to this set cuts the edges $\{1, 4\}$ and $\{2, 3\}$ from the complete graph on 4 nodes, yielding precisely $G = \text{skel}(\mathcal{G})$. This is consistent with Theorem 4.8, as $\mathcal{G} = \mathcal{G}^{tr}_C$ holds for this DAG. However, the subsequent orientation according to the rules in List 1 leads to an incorrect output. For the pair $(1, 4)$, Algorithm 1 sets $\textit{Sepset}(1, 4) = \{3\}$, and applying $\mathcal{R}1$ to the unshielded triple $1 - 2 - 4$ would orient it as $1 \to 2 \leftarrow 4$ due to the fact that $2 \notin \textit{Sepset}(1, 4)$. □

From the point of view of graph separation, the "error" in the previous example occurs because the critical DAG $\mathcal{G}^*_{\{2\}}$ contains the edge $1 \to 4$, which is a $*$-connecting path of type (A). However, 1 and 4 are not $*$-connected in $\mathcal{G}^*_{\{2,3\}}$ and indeed one has that $[1 \perp\!\!\!\perp 4 | 23] \in \text{global}_{C^*}(\mathcal{G}, C)$. So while the node 2 does not lie in the minimal separating set for 1 and 4, 2 does lie in *some* separating set. This suggests that for collider orientation it may be necessary to look beyond the first separating set encountered. The following proposition states that non-minimal separating sets of a certain form provide sufficient information to orient all colliders.

Proposition 4.10. Let $(\mathcal{G}, C)$ *be a weighted DAG and* $\{i, k, j\}$ *be an unshielded triple in* $\text{skel}(\mathcal{G}^{tr}_C)$*. Then the following holds:*

$i \to k \leftarrow j$ *is a collider in* $\mathcal{G}_C^{tr} \iff [i \perp\!\!\!\perp j | k \cup \mathrm{pa}(j) \cup \mathrm{pa}(i)] \notin \mathrm{global}_{C^*}(\mathcal{G}, C)$.

Proof. The direction "⇒" is straightforward: If $i \to k \leftarrow j$ is the orientation of the triple $\{i, k, j\}$ in the true DAG $(\mathcal{G}, C)$, then this is a $*$-connecting path in $\mathcal{G}_K^*$ of type (C) for any conditioning set K with $k \in K$. In particular, $[i \perp\!\!\!\perp j | k \cup \mathrm{pa}(j) \cup \mathrm{pa}(i)] \notin \mathrm{global}_{C^*}(\mathcal{G}, C)$.

We prove "⇐" by contraposition: if $\{i, k, j\}$ is an unshielded triple in $\mathcal{G}_C^{tr}$ which is not oriented $i \to k \leftarrow j$, then conditioning on $K := k \cup \mathrm{pa}(i) \cup \mathrm{pa}(j)$ excludes the possibility of $*$-connecting paths in $\mathcal{G}_{C_K}^{tr*}$ between i and j of any type. The assumption directly implies that no path of type (C) exists. All paths of type (B), (D), (E) are blocked by elements of $\mathrm{pa}(i) \cup \mathrm{pa}(j)$. No $i - j$ path of type (A) exists, as this would imply either $i \to j \in \mathcal{G}_C^{tr}$ or $j \to i \in \mathcal{G}_C^{tr}$, contradicting the assumption that $\{i, j, k\}$ is an unshielded triple in $\mathcal{G}_C^{tr}$. □

Remark 4.11. The process of collecting non-minimal separating sets for each i, j of the form above remains polynomial in the maximal in-degree of $\mathcal{G}$. This is important for the complexity of the modified PC algorithm which will be discussed at the end of this section.

With the slight modifications described above, given $\mathrm{global}_{C^*}(\mathcal{G}, C)$, one can detect and correctly orient all of the unshielded colliders of $\mathcal{G}_C^{tr}$. After subsequently orienting edges according to the rules $\mathcal{R}2 - \mathcal{R}4$ (recall that these are not dependent on the input statements), one obtains an approximation of $\mathcal{G}_C^{tr}$ which is 'at least as good' as the output of the PC algorithm in the d-separation setting, in the sense that the skeleton and unshielded colliders of a graph capable of encoding the same conditional independence information as the true DAG is recovered. Due to the definition of $\perp_{C^*}$ (in particular its reliance on *directed* critical paths), additional edges may be oriented.

4.3. Orienting Induced Cycles

In this section we show how the additional statements which arise from $\perp_{C^*}$ may be used to orient additional edges in the CPDAG which are not orientable in either the $\perp_{\mathrm{d}}$ or $\perp_*$ paradigm. We begin with the following example which shows that graphs which are Markov equivalent with respect to $\sim_*$ / $\sim_{\mathrm{d}}$ may have differing $\perp_{C^*}$ global Markov properties.

Example 4.12. The diamond graph from Example 4.9 and the graph from Example 2.5 are Markov equivalent w.r.t. $\sim_{\mathrm{d}}$ / $\sim_*$, as they have the same undirected skeleton and both contain the unshielded collider $2 \to 4 \leftarrow 3$. However, we have seen in Example 4.9 and Example 4.1 that there exist choices of edge weights such that their $\perp_{C^*}$ global Markov properties differ. (In fact, the $\perp_{C^*}$ global Markov properties of these two graphs differ for any choice of edge weights by Lemma 4.5.) At the same time, their $\perp_{C^*}$ Markov properties encode information about their respective critical paths. For example, in the diamond from Example 4.9 the statement $[1 \perp\!\!\!\perp 4 | 3]$ holds because $1 \to 3 \to 4$ is the unique critical $1 - 4$ path, whereas the fact that $2 \to 1 \to 3 \to 4$ is the unique critical $2 - 4$ path in the graph from Example 4.1 gives rise to the statement $[2 \perp\!\!\!\perp 4 | 13]$. □

Differences such as those highlighted in Example 4.12 allow *induced cycles* of $\mathcal{G}_C^{tr}$ of a certain kind to be oriented.

Definition 4.13. Let $\mathcal{G} = (V, E)$ be a graph. For $W \subset V$, the *induced subgraph* of $\mathcal{G}$ w.r.t W is the subgraph $\mathcal{G}[W]$ with node set W containing the edge $i \to j \in E$ if and only if $i, j \in W$. If $\mathcal{G}[W]$ is a simple cycle, we call it an *induced cycle* of $\mathcal{G}$.

Observe that an induced cycle of a DAG containing exactly one collider has a unique *source* (i.e. a node with induced in-degree zero), which determines the orientation of all other edges. Under certain additional assumptions, the sources of cycles of this type in $\mathcal{G}_C^{tr}$ may be located by querying $\mathrm{global}_{C^*}(\mathcal{G}, C)$.

Lemma 4.14. For a weighted DAG $(\mathcal{G}, C)$ with generic C, let $\mathcal{H} = \mathcal{G}_C^{tr}[W]$ be an induced cycle of $\mathcal{G}_C^{tr}$ with the following two properties:

(i). $\mathcal{H}$ contains a unique unshielded collider $k_1 \to k \leftarrow k_2$,

(ii). For any $w \in W \backslash k$, the unique critical $w - k$ path is contained entirely in W.

For $i \in W \backslash \{k_1, k, k_2\}$ the following holds:

$$i \text{ is the source of } \mathcal{H} \iff \#\{j \in W \mid \exists\, K \subset V \backslash W \text{ s.t } [i \perp_{C^*} k | jK]\} > 0 \text{ is maximal in } W \setminus \{k_1, k, k_2\}. \quad (4.2)$$

Moreover, the maximum is the length of the critical path from the source to the sink k.

Proof. We start by considering the case in which $\mathcal{G}_C^{tr}$ itself is a cycle. In this setting, the statements appearing in the set on the right-hand side of (4.2) have conditioning sets of cardinality one. If $i \in V \backslash \{k, k_1, k_2\}$ is the source then there exist two directed $i - k$ paths

$$\pi_1 \ : \ i = j_{1,0} \to j_{1,1} \to j_{1,2} \to \cdots \to j_{1,m_1} = k_1 \to k$$
$$\pi_2 \ : \ i = j_{2,0} \to j_{2,1} \to j_{2,2} \to \cdots \to j_{2,m_2} = k_2 \to k.$$

Assume that π_1 is the (by genericity of C unique) critical directed $i - k$ path.

Then there exist m_1 statements which separate i and k of the form $[i \perp_{C^*} k | j_{1,\ell}]$ for $\ell \in \{1, \dots m_1\}$. These are the only separation statements of cardinality one separating i and k, as for any intermediate j' along π_2, the critical DAG $\mathcal{G}_{j'}^*$ contains the edge $i \to k$ arising from the unblocked critical path π_1. This edge is a $*-$ connecting path of type (A). Any other $j_{1,\ell}$ along π_1 will be separated from k only given any one of the intermediate nodes $j_{1,\ell+1} \cdots j_{1,m_1}$. This is because for any $q \in \{\ell+1, \dots m_1 - 1\}$, and $p \in \{0, \dots, \ell\}$ the critical DAG $\mathcal{G}_{j_{1,q}}^*$ does not contain the edge $j_{1,p} \to k$, meaning that no $*-$ connecting path between $j_{1,\ell}$ and k of type (A) or (B) is present. The possibility of $*-$ connecting paths of types (C)–(E) is excluded by the fact that the underlying DAG contains only one collider at k. Moreover, none of the nodes along the non-critical path of the form $j_{2,\ell}$, for $\ell \in \{1, \dots, m_2 - 1\}$ are separated from k upon conditioning on one node, as no single node simultaneously blocks π_1 and the directed $i - j_{2,\ell}$ portion of the cycle. More specifically, for any $j \in V \backslash \{k_1, k, k_2, j_{2,\ell}\}$, the critical DAG $\mathcal{G}_j^*$ contains the edge $j_{2,\ell} \to k$ in the case that $j \notin \{j_{2,\ell+1}, \dots j_{2,m_2}\}$ and the 3-node

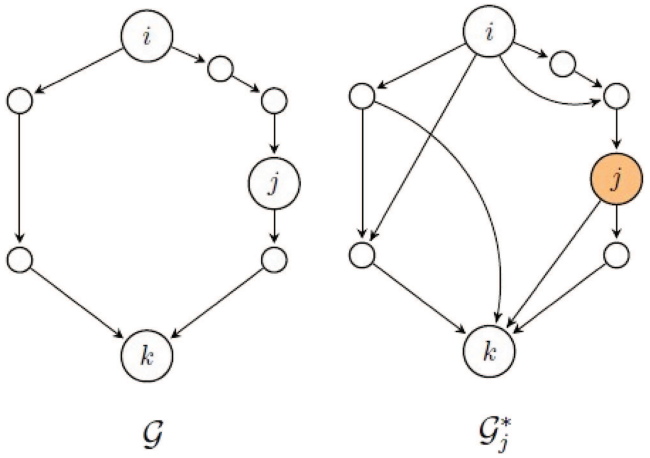


Figure 6. Left: an orientable cycle $\mathcal{G}$. Right: the critical DAG $\mathcal{G}_j^*$, if the weights are chosen such that the directed path through j is the unique $i - k$ critical path.

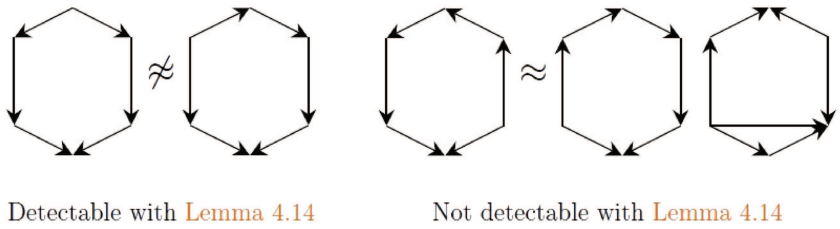


Figure 7. The two cycles on the left have differing global Markov properties. Of the cycles on the right, the first two have equal global Markov properties, meaning that they are indistinguishable at the level of CI data. The cycles of the rightmost graph are not orientable as they are either not induced, or do not contain exactly one unshielded collider.

trek $j_{2,\ell} \leftarrow i \rightarrow k$ otherwise. In either case, $j_{2,\ell}$ and k are $*-$ connected given j. This completes the proof in the case that $\mathcal{G}_C^{tr}$ itself is a cycle (Figure 6).

To generalize the proof for induced cycles it suffices to observe that given a weighted transitive reduction $\mathcal{G}_C^{tr}$ and an induced cycle $\mathcal{H} = \mathcal{G}_C^{tr}[W]$ the following holds:

$$\begin{aligned} &[i \perp_{C^*} k | j] \text{ holds in } (\mathcal{H}, C_{\mathcal{H}}) \\ \Longleftrightarrow &[i \perp_{C^*} k | j \cup \text{an}(W)] \text{ holds in } (\mathcal{G}_C^{tr}, C^{tr}) \,, \end{aligned} \tag{4.3}$$

where $C_{\mathcal{H}}$ and C^{tr} are the restrictions of C to the supports of $\mathcal{H}$ and $\mathcal{G}_C^{tr}$ respectively. This equivalence (4.3) holds because adding $\text{an}(W)$ to the conditioning set blocks all treks from $W \backslash k$ to k containing intermediate nodes outside of W. Furthermore, the second condition of Lemma 4.14 implies that any directed path between nodes in W containing intermediate nodes outside of W does not contribute to the critical path structure of $\mathcal{H}$. Taken together, these last two facts imply that the critical DAG $\mathcal{H}_j^*$ is precisely the restriction of $\mathcal{G}_{C_{j \cup \text{an}(W)}}^{tr*}$ to W, which in turn implies (4.3). This completes the proof. □

We call induced cycles which fulfill the two of conditions (4.14) of Lemma 4.14 *orientable*. The second condition is generally not verifiable given only the global Markov property. However, a sufficient condition for orientability can be verified given the information obtained in the skeleton retrieval and collider orientation steps (Figure 7).

Proposition 4.15. Let $\mathcal{G}_C^{tr}$ be the weighted transitive reduction of some true DAG $(\mathcal{G}, C)$ with C generic, and $\mathcal{H} = \mathcal{G}_C^{tr}[W]$ be an induced cycle of $\mathcal{G}_C^{tr}$ containing exactly one unshielded collider. $\mathcal{H}$ is orientable if it fulfills the following condition:

For all $w \in W \backslash k$ any undirected $w - k$ path containing nodes outside of W contains an unshielded collider in $\mathcal{G}_C^{tr}$.

Proof. One has to show that the condition in Proposition 4.15 implies the second statement of (4.14). This follows immediately from the observation that for all $w \in W \backslash k$ if any $w - k$ path containing nodes outside of W contains a collider triple, then no trek or directed path from w to k exists apart from those which factor entirely through W. □

While Proposition 4.15 is a stronger condition on $\mathcal{G}_C^{tr}$ than the second statement of (4.14), it still allows for the orientability of cycle graphs and the cycles present in certain kinds of fat path' graphs (Fink, Rajchgot, and Sullivant 2016), as well as level-1 phylogenetic networks (Steel 2016).

Remark 4.16. Using an argument similar to that of Lemma 4.14, the set on the right hand side of (4.2) is empty for all $i \in W \backslash \{k_1, k_2, k\}$ if and only if the source of the induced cycle is either k_1 or k_2. However, it is impossible to distinguish between these two cases by querying $\text{global}_{C^*}(\mathcal{G}, C)$; if no $i \in W \backslash \{k_1, k_2, k\}$ is separated from k upon conditioning on only one other node j in the cycle, then there exist two $*-$ connecting paths from i to k in $\mathcal{H}_j^*$, which due to our assumptions must correspond to paths of type (A) and (B) through k_1 and k_2. Thus, if we encounter such a cycle, we can do no better than to mark' k_1 and k_2 as possible sources.

The arguments above rely on the fact that in a cycle containing a unique collider k, each node $i \in W \backslash \{k\}$ is connected to k

Table 1. Average values for the output of `PCstar` on randomly generated weighted DAGs of varying size and sparsity level.

$\|V\|$	a	# Edges of $\mathcal{G}$	# Edges of $\mathcal{G}_C^{tr}$	# Recovered edges (w/o cycle orientation)	# Recovered edges (with cycle orientation)
10	2	9.17	8.66	3.88	4.43
	3	13.13	10.54	6.38	7.54
	4	16.13	11.57	6.99	8.33
	5	19.63	12.15	6.94	8.60
15	2	15.63	14.89	8.53	9.58
	3	17.46	15.95	10.02	11.31
	4	22.74	18.21	12.83	14.59
	5	28.30	19.61	14.17	15.92
20	2	20.36	19.74	11.51	12.47
	3	22.57	21.31	13.74	15.32
	4	28.13	24.38	17.85	20.14

The quantity a is the maximal in-degree of the true DAG $\mathcal{G}$.

either by two directed paths (if i is the source) or one directed path and one trek through the source (otherwise). Thus, they do not generalize to the case of induced cycles containing more than one collider.

Example 4.17. Assume that, given some true weighted DAG $(\mathcal{G}, C)$, the skeleton of its weighted transitive reduction (i.e. the output of Algorithm 1 applied to $\text{global}_{C^*}(\mathcal{G}, C)$) is a hexagon. Due to acyclicity, $\mathcal{G}_C^{tr}$ must contain at least one collider. These colliders may be found with the condition from Proposition 4.10. Consider first the case that there is a unique collider, w.l.o.g $6 \to 5 \leftarrow 4$.

If the source of the cycle is at 6 or 4, then no node in $\mathcal{G}_C^{tr}$ can be separated from the sink 5 by conditioning on one node. In accordance with Remark 4.16, we cannot distinguish from these two cases by querying $\text{global}_{C^*}(\mathcal{G}, C)$. If the source of the cycle is any other node, then we may find it by the condition in Lemma 4.14.

If the hexagon contains two colliders, for example $6 \to 5 \leftarrow 4$ and $1 \to 2 \leftarrow 3$, then the remaining undirected edges $1 - 6$ and $3 - 4$ cannot be oriented. In fact, in this example, each of the four possibilities of orienting these two edges give rise to the same $\perp_{C^*}$ global Markov property. □

We conclude this section by introducing the `PCstar` algorithm, which utilizes the previous results to reconstruct a partially oriented weighted transitive reduction of a true unknown graph given its C^*-global Markov property.

Theorem 4.18. The output of Algorithm 2 is a CPDAG with the same undirected skeleton, unshielded colliders, and orientable induced cycles as $\mathcal{G}_C^{tr}$. Without the optional cycle orientation step, its complexity is $\mathcal{O}(n^{a+2})$, where $n = |V|$ and $a = \max_{v \in V} \text{indeg}(v)$.

Proof. The correctness of the output is a direct consequence of Theorem 4.18, Proposition 4.10, and Lemma 4.14. For the complexity, recall that the worst-case runtime of the PC algorithm is that of Algorithm 1, which lies in $\mathcal{O}(n^{a+2})$, see (Spirtes, Glymour, and Scheines 1993, p.110). To prove that `PCstar` without cycle orientation has the same complexity, it suffices to show that none of the computations in lines 1, 2, and 5 of Algorithm 2 have poorer worst-case performance. Line 1 is exactly Algorithm 1. The complexity of Line 2 is the same as that of the collider orientation step of the original PC algorithm by Remark 4.11, and Line 5 is the same edge orientation step as that of PC. Both of these procedures are polynomial in $|E|$ and in particular bounded above by n^{a+2} (Meek 2023, p. 143). □

The worst-case performance of detecting all orientable induced cycles is $\mathcal{O}(a^n)$; this bound comes from finding all cycles which contain a given unshielded triple. This computation becomes intractable even for relatively small n. However, the actual orientation of an orientable induced cycle with $|W|$ nodes involves querying the set $\text{global}_{C^*}(\mathcal{G}, C)$ a total of $|W|^2$ times. This suggests an alternative approach which replaces lines 3–4 of Algorithm 2 with the orientation of only certain cycles which are selected manually from the output of lines 1–2. For example, to determine the orientation of a single undirected edge of the CPDAG obtained by lines 1–2, one could determine only the induced cycles which contain it and check for orientability by verifying the conditions in Lemma 4.14, but we leave this to future work.

5. Implementation and Outlook

We implemented `PCstar` in julia (Bezanson et al. 2017), using the packages Graphs.jl (Fairbanks et al. 2021) and OSCAR.jl (Decker et al. 2025) for graph and tropical arithmetic functionality respectively. Methods for applying the edge orientation rules were imported from the CausalInference.jl (Schauer 2025) package. The source code may be found at https://github.com/fpnowell/starskeleton.

We ran our function `PCstar` on data produced by randomly generated weighted DAGs on 10, 15, and 20 nodes of varying degrees of sparsity. For the 15 node graphs onwards, we did not generate the entire set of CI statements in the corresponding model, opting instead to query the model as needed by checking for C^*- separation in the true DAG. The runtime of our implementation increases rapidly with increasing n and maximal in-degree a. Despite this, our algorithm terminated in under 24 hours on 31-node graphs of in-degrees 2 and 3; quantities which are motivated by the real-world data set of the Danube hidden river network commonly used in extremal statistics (Asadi, Davison, and Engelke 2015). The results are portrayed in Table 1.

Another question of interest is whether the additional statements implied by C^*-separation can allow for the identifiability of edge directions beyond those which can be oriented by

Algorithm 2 PCstar

Input : A complete set of CI statements $\text{global}_{C^*}(\mathcal{G}, C)$ coming from a graphical model faithful to C^*-separation

Output: A CPDAG with the same skeleton, unshielded colliders, and orientable cycles as the sparsest graph with the same Markov property as $(\mathcal{G}, C)$ for an appropriate choice of weights

Recover $G^{tr} = \text{skel}(\mathcal{G}_C^{tr})$ by applying Algorithm 1 to $\text{global}_{C^*}(\mathcal{G}, C)$

Detect and orient the unshielded colliders of G^{tr} using Proposition 4.10

Optional: **for** *each orientable induced cycle* $\mathcal{G}_C^{tr}[W] = \mathcal{H}$ **do**

    Orient $\mathcal{H}$ using Lemma 4.14

Orient the remaining edges of G^{tr} according to $\mathcal{R}2 - \mathcal{R}4$ of List 1

Lemma 4.14. To answer this one would have to determine if for any DAG $\mathcal{H}$ whose weighted transitive reduction has the same skeleton and oriented edges as the CPDAG returned by `PCstar`, there exists a choice of edge weight matrix $C_{\mathcal{H}}$ such that $\text{global}_{C^*}(\mathcal{H}, C_{\mathcal{H}})$ is the same as the global Markov property of the true DAG $(\mathcal{G}, C)$. While we believe this to be the case for cycles such as the one from Example 4.17, showing this claim for general graphs is a highly non-trivial task. One approach may be to use the polyhedra associated to max-linear CI structures defined in recent work (Boege et al. 2025), but we leave this for future work as we consider it beyond the scope of this paper.

The main obstacle to the real-world application of `PCstar` is the lack of a canonical conditional independence test. While linear SEMs are inherently parametric, meaning that a CI test can be obtained by imposing rank conditions on an estimated covariance matrix, no similar procedure exists for MLBNs. Thus, future work may involve the development of specialized non-parametric conditional independence testing (Shah and Peters 2018) for MLBNs. Other potential avenues include a combination of our methods with recent work on edge weight estimation of MLBNs (Adams, Ferry, and Yoshida 2025) and the development of alternative causal discovery tools for MLBNs based on methods such as greedy equivalence search (Chickering 2003) and 'do'-calculus (Pearl 2009).

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